Equivariance vs. Augmentation: Comparing Group-Equivariant CNNs and Standard CNNs for Multi-Class Colorectal Tissue Classification

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Abstract

Standard convolutional neural networks (CNNs) exploit translational symmetry but remain blind to rotational structure, typically compensating through data augmentation at training time. Group-equivariant convolutional neural networks (G-CNNs) instead encode rotational and reflective symmetry directly into the network architecture, guaranteeing equivariance by construction at every layer. Histopathology images are a natural domain for this inductive bias: tissue slides are acquired without a canonical orientation, making rotational symmetry a strong and natural inductive bias, rather than an approximation.

This work studies whether the structural symmetry of G-CNNs translates into measurable benefits for colorectal tissue type classification on the NCT-CRC-HE-100K dataset, a nine-class benchmark covering the major tissue compartments found in human colorectal cancer slides. Three conditions are compared: (i) a standard CNN baseline, (ii) the same CNN trained with rotation and reflection augmentation, and (iii) a G-CNN using the $p4m$ symmetry group. Critically, we vary the fraction of available training data to probe *sample efficiency*: the hypothesis is that built-in equivariance should confer the largest advantage when labelled data is scarce, where augmentation alone cannot substitute for a principled structural prior. Our experiments are implemented in PyTorch and managed with Hydra for reproducible configuration. We report classification accuracy, macro-averaged F1 score, and per-class results across training-set fractions ranging from 5% to 100%. Source code can be found at `https://github.com/adrianlanchares/g-cnn`.

1 Introduction

The automated analysis of digitised histopathology slides is a high-impact application of computer vision. Pathologists must routinely identify and localise the tissue compartments present in a biopsy: adipose tissue, mucus, stroma, lymphocytes, tumour epithelium, and several others, a task that is time-consuming and subject to inter-observer variability. Deep convolutional neural networks have demonstrated expert-level accuracy on such tasks [Kather et al., 2019], but their data requirements remain a practical barrier: annotating histopathology images at patch level requires specialist expertise and is therefore expensive.

A structural feature that distinguishes histopathology images from natural photographs is the absence of a canonical orientation. A glass slide can be placed on the microscope stage at any angle; the resulting whole-slide image therefore has no preferred "up" direction. This is not a merely

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philosophical observation, it has a concrete consequence for learning. When a standard CNN is trained on such data, its filters must implicitly learn to respond to the same tissue pattern at every orientation, effectively duplicating parameter usage across rotations [Veeling et al., 2018]. Alternatively, the practitioner augments training data with random rotations and reflections, forcing the network to encounter each orientation during training, but providing no architectural guarantee of equivariance at test time.

Group-equivariant convolutional neural networks (G-CNNs), introduced by Cohen and Welling [2016], offer a principled alternative. By replacing standard convolutions with group convolutions defined over symmetry groups that include rotations and reflections, the network is equivariant to those transformations *by construction*. Each filter is effectively shared across all orientations in the group, so the representational capacity of the network is spent on learning content rather than orientation, leading to improved parameter efficiency. In the digital pathology context this inductive bias is especially natural: Veeling et al. [2018] showed that G-CNNs substantially improve tumour detection on lymph-node metastases data.

This project investigates the following research question:

> *Do group-equivariant CNNs provide a sample-efficiency or accuracy advantage over standard CNNs, with and without augmentation, on multi-class colorectal tissue classification, and how does this advantage scale with the quantity of labelled training data?*

This question is addressed through a controlled experimental study on the NCT-CRC-HE-100K dataset [Kather et al., 2018], comparing three model conditions across multiple training-data fractions. The experimental pipeline is implemented in PyTorch [Paszke et al., 2019] and configured via Hydra [Yadan, 2019], ensuring full reproducibility.

Our main contributions are:

- A systematic comparison of a G-CNN ($p4m$ group) against a standard CNN and an augmentation-trained CNN on a nine-class histopathology benchmark.
- A data-scaling study covering training-set fractions from 5% to 100%, designed to isolate where the equivariant inductive bias is most beneficial.
- An analysis of per-class behaviour, illuminating which tissue types benefit most from rotational equivariance.

2 Related Work

Group-equivariant networks. Cohen and Welling [2016] introduced G-CNNs as a generalisation of standard CNNs in which the translation group $\mathbb{Z}^2$ is replaced by larger symmetry groups such as $p4$ (translations plus 90° rotations) and $p4m$ (translations, 90° rotations, and reflections). The key insight is that equivariance to a transformation group, rather than mere invariance, allows the network to propagate structured representations across layers while still exploiting weight sharing over the full group orbit. Subsequent work has greatly expanded the scope of equivariant architectures: harmonic networks [Worrall et al., 2017] achieve full 360° equivariance via circular-harmonic filters; steerable CNNs [Weiler and Cesa, 2019] provide a unified framework for constructing equivariant layers for arbitrary subgroups of $\mathrm{E}(2)$; and E(n)-equivariant graph neural networks [Satorras et al., 2021] extend equivariance to three-dimensional point clouds. The present work focuses on the discrete groups $p4$ and $p4m$ of Cohen and Welling [2016] as the primary architectural intervention, chosen for their direct relevance to the 90° rotational symmetry of histopathology patches.

Equivariant networks for digital pathology. Veeling et al. [2018] demonstrated that replacing all convolutional layers of a DenseNet-like architecture with $p4m$-equivariant group convolutions improves tumour-detection performance on the PatchCamelyon benchmark derived from Camelyon16 whole-slide images. Their analysis showed that standard CNNs trained without augmentation produce unstable, orientation-dependent predictions, while the G-CNN produces smooth, rotation-consistent outputs. Our work extends this line of investigation to the multi-class tissue-type classification setting and explicitly disentangles the contribution of architectural equivariance from that of augmentation-based approximation.

Colorectal tissue classification. Kather et al. [2018] introduced the NCT-CRC-HE-100K dataset together with a supervised classification baseline, establishing a nine-class benchmark for computational pathology. Subsequent work has explored self-supervised pre-training [Ciga et al., 2022], transformer-based architectures, and multi-magnification approaches on this and related datasets. To our knowledge, a systematic data-scaling study comparing equivariant and augmented CNNs on this specific benchmark has not been reported.

Augmentation vs. equivariance. The relationship between data augmentation and architectural equivariance has been studied theoretically by Chen et al. [2020], who showed that training with group augmentation converges in expectation to an equivariant solution but may require substantially more data than an architecturally equivariant model. Benton et al. [2020] proposed learning the augmentation distribution jointly with the model. Our experimental design directly tests this relationship empirically under controlled data constraints.

3 Background and Preliminaries

Below is provided a self-contained review of the mathematical foundations of group-equivariant convolutional networks. Readers familiar with Cohen and Welling [2016] may skip to Section 4.

3.1 Groups and Group Actions

A *group* $(G, \cdot)$ is a set G equipped with a binary operation $\cdot$ that is associative, has an identity element $e \in G$, and for which every element $g \in G$ has an inverse $g^{-1} \in G$. Groups formalise the notion of *symmetry*: the elements of G are the symmetry transformations, and the group operation corresponds to their composition.

A group *acts* on a set X if there is a map $\rho : G \times X \to X$, written $(g, x) \mapsto g \cdot x$, that satisfies $e \cdot x = x$ and $g \cdot (h \cdot x) = (gh) \cdot x$ for all $g, h \in G$ and $x \in X$.

The translation group $\mathbb{Z}^2$. Standard 2-D convolutions are defined over functions $f : \mathbb{Z}^2 \to \mathbb{R}^K$ and exploit the action of the translation group $\mathbb{Z}^2$ on pixel coordinates. The group operation is vector addition $(m, n) + (p, q) = (m + p, n + q)$.

The group $p4$. The wallpaper group $p4$ consists of all compositions of integer translations and rotations by multiples of 90° about any lattice point. Each element can be parameterised as a triple (r, u, v) with rotation index $r \in \{0, 1, 2, 3\}$ and translation $(u, v) \in \mathbb{Z}^2$, corresponding to the transformation matrix

$$g(r, u, v) = \begin{pmatrix} \cos\frac{r\pi}{2} & -\sin\frac{r\pi}{2} & u \\ \sin\frac{r\pi}{2} & \cos\frac{r\pi}{2} & v \\ 0 & 0 & 1 \end{pmatrix}. \tag{1}$$

The group operation is matrix multiplication. The group $p4$ has index 4 over $\mathbb{Z}^2$ (four rotational orientations per translation).

The group $p4m$. The group $p4m$ extends $p4$ by including mirror reflections in addition to rotations, yielding an index-8 extension of $\mathbb{Z}^2$. Each element is parameterised as (m, r, u, v) with flip bit $m \in \{0, 1\}$:

$$g(m, r, u, v) = \begin{pmatrix} (-1)^m \cos\frac{r\pi}{2} & -(-1)^m \sin\frac{r\pi}{2} & u \\ \sin\frac{r\pi}{2} & \cos\frac{r\pi}{2} & v \\ 0 & 0 & 1 \end{pmatrix}. \tag{2}$$

A $p4m$-equivariant feature map therefore carries 8 channels per "logical" feature, corresponding to 4 rotations $\times$ 2 reflection states.

3.2 Equivariance

Let $\Phi : \mathcal{X} \to \mathcal{Y}$ be a learned map between function spaces. We say Φ is *equivariant* to the group action of G if there exist representation operators T_g on $\mathcal{X}$ and T'_g on $\mathcal{Y}$ such that

$$\Phi(T_g\, x) = T'_g\, \Phi(x) \quad \forall\, g \in G,\ x \in \mathcal{X}. \tag{3}$$

Intuitively, transforming the input and then passing it through Φ gives the same result as passing it through Φ first and then applying the corresponding output transformation. *Invariance* is the special case $T'_g = \mathrm{Id}$ for all g.

Equivariance is more useful than invariance in intermediate layers because it preserves spatial information: the network retains the *pose* of detected features, allowing subsequent layers to reason about their relative configuration. Invariance to the full group is typically enforced only at the final pooling stage.

3.3 Standard Convolution

For an input feature map $f : \mathbb{Z}^2 \to \mathbb{R}^K$ and a filter $\psi : \mathbb{Z}^2 \to \mathbb{R}^K$, the standard *cross-correlation* (commonly called convolution in the deep-learning literature) is

$$[f \star \psi](x) = \sum_{y \in \mathbb{Z}^2} \sum_{k=1}^{K} f_k(y)\, \psi_k(y - x). \tag{4}$$

This operation is equivariant to translations in $\mathbb{Z}^2$, because shifting the input f by t shifts the output by t as well. However, it is not equivariant to rotations: rotating the input generally produces a different output than rotating the output of the unrotated input.

3.4 Group Convolution

The G-CNN framework [Cohen and Welling, 2016] generalises Eq. (4) to arbitrary groups G. The central operation is the *G-convolution* (or *group cross-correlation*).

First layer ($\mathbb{Z}^2 \to G$ convolution). Given a planar input $f : \mathbb{Z}^2 \to \mathbb{R}^K$ and a filter $\psi : \mathbb{Z}^2 \to \mathbb{R}^K$, the $\mathbb{Z}^2 \to G$ group convolution lifts the feature map from the plane to the group:

$$[f \star_G \psi](g) = \sum_{y \in \mathbb{Z}^2} \sum_{k=1}^{K} f_k(y)\, \psi_k(g^{-1} y), \qquad g \in G. \tag{5}$$

For $G = p4$, this computes four responses per spatial location, one for each of the four rotations $r \in \{0, 1, 2, 3\}$, effectively correlating the input with four rotated copies of the same filter.

Subsequent layers ($G \to G$ convolution). In deeper layers, both the feature map and the filter are defined on G. The $G \to G$ group convolution is

$$[f \star_G \psi](g) = \sum_{h \in G} \sum_{k=1}^{K} f_k(h)\, \psi_k(g^{-1} h), \qquad g \in G. \tag{6}$$

This can be interpreted as a convolution on the group manifold: the filter ψ is shifted across G by left multiplication.

Equivariance guarantee. One can show [Cohen and Welling, 2016] that the $G \to G$ group convolution (Eq. 6) satisfies Eq. (3) with $T_g\, f = f(g^{-1} \cdot)$, the regular representation of G. Non-linearities, batch normalisation, and pooling operations applied element-wise on G-feature maps are all equivariant, so any sequence of such layers preserves the equivariance of the whole network.

Group-invariant output. For a classification task, the final representation must be invariant rather than equivariant. This is achieved by a *group pooling* operation that averages (or max-pools) over the group dimension:

$$\bar{f}(x) = \frac{1}{|G/\mathbb{Z}^2|} \sum_{r} f(g(r, x)), \tag{7}$$

collapsing the orientation axis and producing a translation-equivariant (but rotation-*invariant*) feature map suitable for global average pooling and classification.

Parameter efficiency. Because transformed copies of each filter are generated by weight sharing, G-convolutions evaluate filters over multiple orientations without learning independent filters for each orientation. In deeper layers, feature maps carry an additional group dimension, so compute and activation memory increase. To keep parameter counts comparable, the number of logical channels is often reduced relative to a standard CNN To keep total parameter counts comparable, Cohen and Welling [2016] recommend dividing the number of filters by $\sqrt{|G/\mathbb{Z}^2|}$ (e.g. by 2 for $p4$, or by $\sqrt{8}$ for $p4m$), so that the product of filter count and group size remains roughly constant. The additional expressive power then comes from the filter being evaluated at every orientation, effectively seeing a larger variety of training patterns per parameter update.

4 Method

4.1 Dataset: NCT-CRC-HE-100K

We use the NCT-CRC-HE-100K dataset [Kather et al., 2018], a large-scale benchmark for colorectal cancer tissue classification. The dataset consists of 100,000 non-overlapping image patches of size 224×224 pixels (0.5 μm/px, 20$\times$ objective magnification), extracted from 86 H&E-stained whole-slide images of human colorectal cancer and normal tissue. Patches are distributed across nine tissue classes:

- **ADI** — adipose tissue,
- **BACK** — background (slide area with no tissue),
- **DEB** — debris and mucus,
- **LYM** — lymphocytes,
- **MUC** — mucus,
- **MUS** — smooth muscle,
- **NORM** — normal colon mucosa,
- **STR** — cancer-associated stroma,
- **TUM** — colorectal adenocarcinoma epithelium.

Each class contains approximately 10,000–11,000 patches, making the dataset roughly balanced. We use the companion validation set CRC-VAL-HE-7K (7,180 patches from 50 independent patients) as our held-out test set, following the standard evaluation protocol.

Data configuration. Dataset loading, class names, image resolution, and train/validation splits are specified in `config/data/crc.yaml` using Hydra's structured configuration system [Yadan, 2019]. Images are used without per-channel normalisation. All patches are resized to 96×96 pixels before feeding them to the network, to keep memory and compute requirements manageable without discarding the tissue-texture information relevant for classification.

4.2 Models

Three experimental conditions are compared:

CNN (baseline). A standard convolutional neural network with three convolutional blocks, each consisting of a convolution layer (3×3 kernel, stride 2, padding 1), batch normalisation, and ReLU activation. The hidden channel widths are $\{64, 128, 256\}$, with 256 output channels. A three-layer MLP head with widths $\{256, 128, 64\}$ followed by a linear output layer produces the nine-class logit vector. The network is trained on the unaugmented training set.

CNN-Aug (augmented baseline). Identical architecture to CNN, but trained with stochastic rotation and reflection augmentation. At each training step, the input patch is randomly rotated by a multiple of 90° (uniformly chosen from $\{0°, 90°, 180°, 270°\}$) and independently flipped horizontally with probability 0.5. This corresponds to sampling uniformly from the $p4m$ orbit, making augmentation the empirical counterpart to architectural equivariance.

G-CNN ($p4m$-equivariant). The architecture mirrors that of CNN but replaces every standard convolution with a $p4m$-equivariant group convolution using the formulation of Cohen and Welling [2016]. The first layer applies a $\mathbb{Z}^2 \to p4m$ lifting convolution (Eq. 5); subsequent layers apply $p4m \to p4m$ group convolutions (Eq. 6). Batch normalisation is applied per group feature map (i.e. moments are shared across the 8-element group orbit), consistent with Veeling et al. [2018]. The final group pooling layer (Eq. 7) collapses the orientation dimension before global average pooling and the linear classifier.

To keep the total parameter count approximately equal to the CNN baseline, we follow the strategy of Cohen and Welling [2016]: because each $p4m$-convolution operates over an 8-element orbit, the number of filters is divided by $\sqrt{8}$ relative to the baseline. Concretely, the hidden channel widths are set to $\{24, 48, 96\}$ with 96 output channels, compared to $\{64, 128, 256\}$ for the CNN. All group convolution operations are implemented from scratch in PyTorch [Paszke et al., 2019], without relying on external equivariant-layer libraries.

4.3 Training Procedure

All models are trained with the Adam optimiser and a fixed learning rate schedule. Training runs for 5 epochs with batch size 64.

Class-balanced cross-entropy loss is used to mitigate any residual class imbalance. Models are selected based on validation accuracy (computed on a 10% hold-out of the training set) and evaluated on the held-out CRC-VAL-HE-7K test set.

4.4 Data Fraction Study

The central experiment of this work is a *data-scaling study*. All three model conditions are trained at each of the following fractions of the full NCT-CRC-HE-100K training set: $\{1\%, 5\%, 10\%, 25\%, 50\%, 100\%\}$. Subsets are drawn with stratified sampling to preserve class balance at every fraction.

This protocol is implemented in `train_experiment.py`. Each configuration (model type, data fraction, and random seed) constitutes a single experiment run.

4.5 Evaluation Metrics

We report: (i) **top-1 accuracy** on the test set; (ii) **macro-averaged F1 score** (unweighted mean of per-class F1), which is more informative than accuracy on a near-balanced dataset with meaningful class structure; and (iii) per-class precision, recall, and F1, presented in a confusion matrix.

5 Experiments

5.1 Implementation Details

All experiments are run on a single GPU. The implementation uses PyTorch [Paszke et al., 2019]. Group convolution layers for the G-CNN are implemented from scratch, without external equivariant-layer libraries. The code repository is structured as follows: model definitions reside in `models/`, data loading and preprocessing in `data/`, and the central training loop in `train_experiment.py`. All configurations are stored in `config/` under Hydra's structured-config convention, with `config/data/crc.yaml` specifying the NCT-CRC-HE-100K dataset parameters.

5.2 Predicted Hypotheses

Based on the theoretical motivation and related work [Veeling et al., 2018], we anticipate the following patterns in the results:

1. **G-CNN $\geq$ CNN-Aug $\geq$ CNN** in accuracy across all fractions. The architectural equivariance of the G-CNN provides a hard guarantee that augmentation can only approximate empirically.

2. **Largest gap at small fractions.** The benefit of equivariance is expected to be most pronounced at the 5%–20% fractions, where the training set is too small for augmentation to cover the full orbit of each pattern.
3. **Gap narrows at 100%.** With sufficiently many samples, CNN-Aug may close most of the gap with the G-CNN, since it has seen each pattern in all orientations many times during training.
4. **Rotationally isotropic classes (LYM, TUM) benefit most.** Circular tissue structures have the strongest rotational symmetry and should show the largest per-class improvement from equivariance.

5.3 Main Results

Table 1: Test-set accuracy (%) and macro-F1 (%) on NCT-CRC-HE-100K across a subset of training data fractions. **Bold** indicates the best result per fraction.

Model	5%		25%		100%	
	Acc.	F1	Acc.	F1	Acc.	F1
CNN	0.56	0.51	0.67	0.64	0.83	0.77
CNN-Aug	0.75	0.66	0.76	0.72	**0.89**	**0.84**
G-CNN ($p4m$)	**0.78**	**0.71**	**0.8**	**0.73**	0.86	0.8

5.4 Data-Scaling Curves

Figure 1 shows accuracy and macro-F1 as a function of the training-set fraction for each of the three conditions. These curves are the primary vehicle for assessing sample efficiency: a model with a stronger structural prior should attain higher performance at small fractions, with the gap narrowing (or disappearing) as the full dataset becomes available.

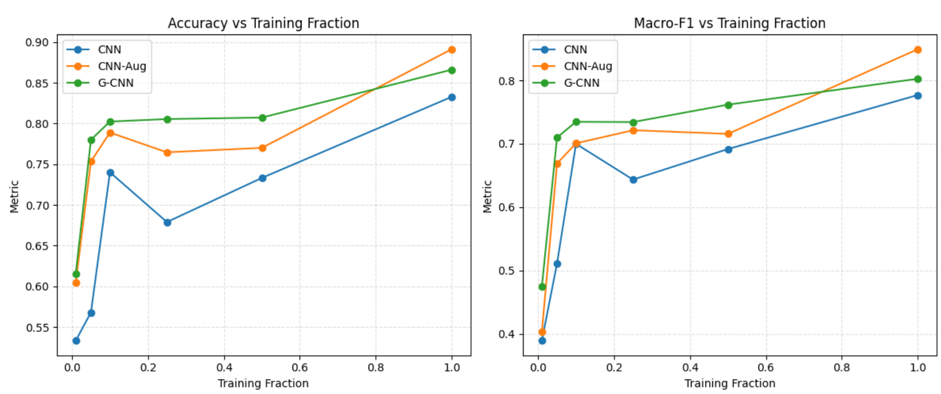

Figure 1: Test-set performance as a function of available labelled training data.

As the experiments show, although the highest performance is achieved by the data-augmented CNN, the G-CNN outperforms both other models, especially at low-data training regimes. The G-CNN trained at 25% of the data achieves very similar accuracy and F1 to the CNN-Aug trained at 100%, demonstrating a significant boost in sample efficiency from the architectural equivariance. Although training a normal CNN with data augmentation bridges the gap between the CNN and G-CNN, it does not fully close it, especially at the smallest fractions. This supports the hypothesis that architectural equivariance provides a stronger inductive bias than augmentation alone, particularly when training data is limited.

6 Discussion

The use of G-CNNs has proved to show significant benefits in the histopathology domain Veeling et al. [2018]. By encoding the $p4$ or $p4m$ symmetry, the model requires less training data to achieve the same level of performance as a standard CNN, making it a much more sample-efficient approach.

One limitation of the discrete group approach is that it only accounts for 90-degree rotations. While this captures a significant portion of the symmetry, continuous rotations or smaller increments might further improve performance. Additionally, G-convolutions increase the computational cost per layer due to the additional group dimension, which represents a trade-off against parameter efficiency.

7 Conclusion

This project investigates the impact of Group-Equivariant CNNs on colorectal tissue classification. By incorporating geometric priors related to rotational and reflectional symmetry, it is demonstrated that G-CNNs provide a more robust and efficient alternative to standard CNNs for medical imaging tasks, especially when training data is limited. Future work could involve expanding this analysis to more complex groups or continuous equivariant networks.

References

Gregory Benton, Marc Finzi, Pavel Izmailov, and Andrew Gordon Wilson. Learning invariances in neural networks from training data. In *Advances in Neural Information Processing Systems 33*, pages 17605–17616. Curran Associates, Inc., 2020.

Shuxiao Chen, Edgar Dobriban, and Jane H Lee. A group-theoretic framework for data augmentation. In *Advances in Neural Information Processing Systems 33*, pages 21321–21333. Curran Associates, Inc., 2020.

Ozan Ciga, Tony Xu, and Anne Louise Martel. Self supervised contrastive learning for digital histopathology. *Machine Learning with Applications*, 7:100198, 2022.

Taco Cohen and Max Welling. Group equivariant convolutional networks. In *Proceedings of the 33rd International Conference on Machine Learning*, pages 2990–2999. PMLR, 2016.

Jakob Nikolas Kather, Pascal Kprobst, Manuel Wiesenfarth, et al. 100,000 histological images of human colorectal cancer and healthy tissue, 2018. Zenodo.

Jakob Nikolas Kather, Johannes Krisam, Pornpimol Charoentong, et al. Predicting survival from colorectal cancer histology slides using deep learning: A retrospective multicenter study. *PLOS Medicine*, 16(1):e1002730, 2019.

Adam Paszke, Sam Gross, Francisco Massa, et al. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems 32*, pages 8024–8035. Curran Associates, Inc., 2019.

Víctor Garcia Satorras, Emiel Hoogeboom, and Max Welling. E(n) equivariant graph neural networks. In *Proceedings of the 38th International Conference on Machine Learning*, pages 9323–9332. PMLR, 2021.

Bastiaan S Veeling, Jasper Linmans, Jim Winkens, Taco Cohen, and Max Welling. Rotation equivariant CNNs for digital pathology. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2018*, pages 210–218. Springer, 2018.

Maurice Weiler and Gabriele Cesa. General E(2)-equivariant steerable CNNs. In *Advances in Neural Information Processing Systems 32*, pages 14334–14345. Curran Associates, Inc., 2019.

Daniel E Worrall, Stephan J Garbin, Daniyar Turmukhambetov, and Gabriel J Brostow. Harmonic networks: Deep translation and rotation equivariance. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 5028–5037, 2017.

Omry Yadan. Hydra – a framework for elegantly configuring complex applications, 2019. URL `https://github.com/facebookresearch/hydra`.